

BOOK MARKET ANALYSIS AND RATING PREDICTION USING WEB SCRAPING AND MACHINE LEARNING

Final Project Report : Data Science PGC Capstone Project

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Tools Used: Python, Jupyter Notebook, Pandas, BeautifulSoup, Requests, Matplotlib, Seaborn, Scikit-learn, Joblib, Microsoft Power BI

Dataset Source: Books to Scrape

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1. EXECUTIVE SUMMARY

This project presents an end-to-end data science workflow for analysing book catalogue data collected through web scraping and applying machine learning to predict book ratings. The project uses the **Books to Scrape** website as the data source. The website provides a suitable sandbox environment for practising web scraping and contains book information such as title, price, rating, availability, category, UPC, stock count, tax, and description. The project was completed through several stages. First, book information was collected from multiple catalogue pages using Python, Requests, and BeautifulSoup. Detailed product pages were also accessed to collect additional attributes. A total of **1,000 book records** were collected and stored as a raw CSV dataset.

The raw dataset was then cleaned and preprocessed using Pandas. Missing descriptions were handled, currency symbols were removed from price and tax fields, ratings were converted from words into numerical values, availability values were standardized, text fields were cleaned, and additional features such as price category, rating category, and description length were created. The final cleaned dataset contained **1,000 records and 13 columns** with no missing values or duplicate records.

Exploratory Data Analysis was then performed to understand the distribution of prices, ratings, categories, stock levels, and description lengths. The analysis showed that the average book price was approximately **£35.07**, the average rating was **2.92 out of 5**, and the average stock count was approximately **8.59**. The catalogue contained **50 categories**.

For machine learning, the project used a regression approach to predict book ratings. Two models were evaluated: **Random Forest Regression** and **Linear Regression**. Random Forest achieved an RMSE of **1.5236** and an R^2 score of **-0.1328**, while Linear Regression achieved an RMSE of **1.5244** and an R^2 score of **-0.1341**. Random Forest was therefore selected as the final model because it performed slightly better.

However, the negative R^2 score indicates that the available features — price, stock count, description length, and category — have limited ability to predict book ratings. This is consistent with the EDA, which found very weak relationships between the numerical variables.

Finally, the cleaned data was imported into Microsoft Power BI to create an interactive dashboard. The dashboard includes KPI cards, charts, slicers, AI visuals, a decomposition tree, bookmarks, drill-through functionality, synchronized slicers, and row-level security. Overall, the project demonstrates a complete data science pipeline from web data collection to cleaning, analysis, machine learning, and interactive business intelligence reporting.

2. PROJECT OBJECTIVE

2.1 Main Objective

The main objective of this project is to collect book catalogue data from a public website, clean and analyse the data, identify useful patterns, build a machine learning model for book rating prediction, and present the results through an interactive Power BI dashboard.

2.2 Specific Objectives

The project aims to:

- Collect book information from multiple web pages using Python web scraping techniques.
 - Build a structured dataset containing at least 1,000 book records.
 - Clean and preprocess the collected data.
 - Handle missing values, duplicates, inconsistent formats, and data types.
 - Create useful derived features for analysis.
 - Perform exploratory data analysis using statistical summaries and visualizations.
 - Investigate relationships between book price, ratings, stock, categories, and descriptions.
 - Build regression models to predict book ratings.
 - Compare machine learning model performance using RMSE and R^2 .
 - Select and save the best-performing model.
 - Create an interactive Power BI dashboard.
 - Provide analytical insights and practical recommendations.
 - Demonstrate the complete end-to-end data science workflow.
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3. PROJECT SCOPE AND METHODOLOGY

The project follows an end-to-end data science workflow consisting of six main phases.

Phase 1 – Web Scraping

Book data was collected from the Books to Scrape website using Requests and BeautifulSoup.

Phase 2 – Data Cleaning

The collected data was cleaned, standardized, validated, and enhanced with additional features.

Phase 3 – Exploratory Data Analysis

Statistical analysis and visualizations were used to identify patterns and relationships in the dataset.

Phase 4 – Machine Learning

Regression models were trained to predict book ratings using selected book attributes.

Phase 5 – Dashboard Development

Microsoft Power BI was used to create an interactive dashboard for analysing the dataset.

Phase 6 – Final Reporting

The complete project findings, model results, dashboard, recommendations, and limitations are documented in this report.

4. PHASE 1 – DATA COLLECTION AND WEB SCRAPING

4.1 Data Source

The project uses **Books to Scrape** as the source website.

The website provides a catalogue of books with multiple pages of products. It was selected because it provides structured information suitable for practising web scraping without requiring user login or authentication.

The scraping process collected information from catalogue pages as well as individual book detail pages.

4.2 Web Scraping Technologies

The following Python libraries were used:

- **Requests** – to send HTTP requests to web pages.
- **BeautifulSoup** – to parse HTML content.
- **Pandas** – to create and manage the dataset.
- **Regular Expressions (re)** – to extract numerical stock information.
- **Time** – to introduce a short delay between requests.
- **urllib.parse.urljoin** – to construct complete product URLs.

4.3 Data Collection Process

The website was first accessed using the Requests library. The HTTP response was then parsed using BeautifulSoup.

The scraper identified individual book records from the catalogue pages using the HTML structure of the website.

For each book, the following basic information was collected:

- Title
- Price
- Rating
- Availability
- Product URL

The scraper then visited each individual book page to collect additional information.

4.4 Detailed Attributes Collected

The final raw dataset contained the following 10 fields:

Field	Description
title	Name of the book
price	Listed book price
rating	Book rating
availability	Availability information
product_url	URL of the individual book page
category	Book category
upc	Universal Product Code
stock_count	Number of books available
tax	Listed tax value
description	Book description

4.5 Pagination

The scraper iterated through the catalogue pages using the website's pagination structure. The complete catalogue was processed rather than collecting information from only a single page.

This resulted in a total of:

1,000 book records

The data was saved as:

raw_books_data.csv

4.6 Initial Data Validation

After scraping, the dataset was converted into a Pandas DataFrame.

The initial dataset contained:

- **1,000 rows**
- **10 columns**

The column names and missing values were checked before moving to the cleaning phase.

4.7 Web Scraping Outcome

The web scraping phase successfully produced the required raw dataset containing 1,000 book records and multiple useful attributes.

The resulting raw dataset was used as the input for Phase 2.

5. PHASE 2 – DATA CLEANING AND PREPROCESSING

5.1 Objective

The objective of the cleaning phase was to convert the raw scraped data into a consistent, validated, analysis-ready dataset.

The main operations included:

- Missing-value handling
 - Duplicate checking
 - Currency cleaning
 - Rating conversion
 - Tax cleaning
 - Availability standardization
 - Text standardization
 - Numerical validation
 - Feature engineering
-

5.2 Loading the Raw Dataset

The raw dataset raw_books_data.csv was loaded into Pandas.

The dataset contained:

1,000 rows × 10 columns

5.3 Missing Values

Missing values were checked across all columns.

The description field contained missing values.

These missing descriptions were replaced with:

"No description available"

This ensured that the description column did not contain missing values during further processing.

5.4 Duplicate Records

Duplicate records were checked using Pandas.

The final dataset contained:

0 duplicate rows

Therefore, no duplicate removal was required.

5.5 Price Cleaning

The raw price values contained currency symbols and formatting characters.

The cleaning process:

1. Removed £ and Â£.
2. Removed unnecessary whitespace.
3. Converted the values to floating-point numbers.

After cleaning, the price field became a numerical variable suitable for statistical analysis and machine learning.

5.6 Rating Conversion

The original ratings were represented as words such as:

- One
- Two
- Three
- Four
- Five

These were converted into numerical values:

Original Rating Numerical Rating

One	1
Two	2
Three	3
Four	4
Five	5

This conversion allowed ratings to be used in numerical analysis and regression modelling.

5.7 Tax Cleaning

The tax field was cleaned by removing currency formatting and converting the values into a numerical data type.

The resulting tax values were all **£0.00**, making tax a constant variable with no useful variation for analysis or machine learning.

5.8 Availability Standardization

Availability information was standardized into a consistent categorical value.

The resulting dataset used:

In Stock

for the available books.

5.9 Numerical Validation

The numerical variables were checked for invalid values.

The validation included:

- Negative prices
- Ratings outside the 1–5 range
- Negative stock counts

No invalid numerical values were identified.

5.10 Feature Engineering

Three additional features were created.

Price Category

Books were grouped according to price:

- **Budget:** Price < £20
- **Mid-Range:** £20–£39.99
- **Premium:** £40 and above

Distribution:

Price Category Number of Books

Budget	196
Mid-Range	401
Premium	403

Rating Category

Ratings were grouped into:

- **Low Rated:** Rating ≤ 2
- **Average Rated:** Rating = 3
- **Highly Rated:** Rating > 3

Distribution:

Rating Category Number of Books

Low Rated	422
Average Rated	203
Highly Rated	375

Description Length

The length of each book description was calculated in characters.

The average description length was approximately:

1,441 characters

5.11 Final Data Quality Check

After cleaning and feature engineering:

- Missing values: **0**
- Duplicate records: **0**
- Final rows: **1,000**
- Final columns: **13**

The final dataset was saved as:

cleaned_books_data.csv

The cleaned dataset was then used for EDA and machine learning.

6. PHASE 3 – EXPLORATORY DATA ANALYSIS

6.1 Objective

The objective of the EDA phase was to understand the structure and characteristics of the cleaned dataset and identify important patterns, distributions, relationships, and anomalies.

The analysis used:

- Summary statistics
- Histograms
- Bar charts
- Scatter plots
- Correlation analysis
- Correlation heatmap
- Boxplots
- Category analysis

6.2 Dataset Overview

The cleaned dataset contains:

- **1,000 books**
- **13 variables**
- **50 categories**

The main numerical variables include:

- Price
- Rating
- Stock count
- Tax
- Description length

6.3 Summary Statistics

Variable	Mean	Minimum	Median	Maximum
Price	£35.07	£10.00	£35.98	£59.99
Rating	2.92	1	3	5
Stock Count	8.59	1	7	22
Tax	0.00	0	0	0
Description Length	1,441.50	24	1,353.50	8,646

6.4 Rating Distribution

The rating distribution was:

Rating Number of Books

1 Star 226
2 Stars 196
3 Stars 203
4 Stars 179
5 Stars 196

The average rating across the catalogue was:

2.92 out of 5

The distribution shows that all five rating levels are represented in the dataset.

6.5 Price Distribution

Book prices ranged from:

£10.00 to £59.99

The average price was:

£35.07

The price distribution covers a relatively broad range, with books spread across Budget, Mid-Range, and Premium segments.

6.6 Price Category Distribution

The dataset contained:

- **196 Budget books**
- **401 Mid-Range books**
- **403 Premium books**

Premium books therefore represented the largest price segment, followed closely by Mid-Range books.

6.7 Average Price by Rating

The average prices by rating were:

Rating Average Price

1	£34.56
2	£34.81
3	£34.69
4	£36.09
5	£35.37

The average prices remain relatively close across the rating groups.

This suggests that there is no strong price difference associated with rating level.

6.8 Average Price by Category

The categories with the highest average prices included:

Category	Average Price
Suspense	£58.33
Novels	£54.81
Politics	£53.61
Health	£51.45
New Adult	£46.38
Christian	£42.50
Sports and Games	£41.17
Self Help	£40.62
Travel	£39.79
Fantasy	£39.59

Suspense had the highest average catalogue price among the categories analysed.

6.9 Price and Rating Relationship

A scatter plot was used to investigate the relationship between book price and rating.

The analysis did not reveal an obvious strong relationship.

This finding is supported by the correlation analysis.

The correlation between:

Price and Rating = approximately 0.028

This represents a very weak positive relationship.

Therefore, higher-priced books cannot be assumed to have higher ratings in this dataset.

6.10 Correlation Analysis

The correlation analysis showed generally weak relationships between the numerical variables.

Important correlations included:

Variables	Correlation
Price – Rating	0.028
Price – Stock Count	-0.011
Price – Description Length	0.022
Rating – Stock Count	0.016
Rating – Description Length	0.025
Stock Count – Description Length	0.056

All of these relationships are weak.

This finding became particularly important during the machine learning phase because the selected features do not have strong relationships with the rating target.

6.11 Price Outlier Analysis

A boxplot was used to investigate potential price outliers.

The book prices remained within the overall range of approximately:

£10–£60

No major extreme price outliers were identified.

6.12 Average Rating by Category

The categories with the highest average ratings included:

Category	Average Rating
Erotica	5.00
Adult Fiction	5.00
Novels	5.00
Christian Fiction	4.17
Health	3.75
Art	3.63
Poetry	3.53
Humor	3.40
Spirituality	3.33
Young Adult	3.30

These results should be interpreted carefully because some categories contain relatively few books. A high average based on a small number of observations may not represent the broader catalogue.

6.13 Category Representation

The most frequently represented categories included:

Category	Number of Books
Default	152
Nonfiction	110
Sequential Art	75
Add a comment	67
Fiction	65
Young Adult	54
Fantasy	48
Romance	35
Mystery	32
Food and Drink	30

Default was the most frequently represented category.

6.14 Stock Analysis

The average stock count was:

8.59 books

Average stock by rating:

Rating Average Stock

1	8.36
2	8.77
3	8.36
4	8.88
5	8.62

The differences are small, indicating no strong relationship between rating and stock availability.

6.15 Description Length Analysis

The average description length was:

1,441.50 characters

Average description length by rating:

Rating Average Description Length

1	1,470.04
2	1,380.40
3	1,440.99
4	1,364.29
5	1,540.71

There is no clear monotonic relationship between description length and rating.

6.16 Key EDA Findings

The main findings from the EDA were:

1. The dataset contains **1,000 books across 50 categories**.
2. The average book price is **£35.07**.

3. Book prices range from **£10.00 to £59.99**.
4. The average rating is **2.92 out of 5**.
5. The average stock count is **8.59**.
6. The average description length is approximately **1,441 characters**.
7. Premium books have a substantially higher average price than Budget and Mid-Range books.
8. Suspense has the highest average catalogue price.
9. The relationship between price and rating is very weak.
10. Numerical variables generally show weak correlations with one another.
11. No major extreme price outliers were observed.
12. Some categories have very high average ratings, but these results should be interpreted carefully because category sizes vary.
13. There is no meaningful time trend because the dataset does not contain a date or time field.

7. PHASE 4 – MACHINE LEARNING

7.1 Machine Learning Objective

The machine learning objective was to develop a regression model capable of predicting book ratings using available book attributes.

The problem was therefore defined as:

Regression – Predict Book Rating

7.2 Features and Target

The following variables were selected as input features:

- price
- stock_count
- description_length
- category

The target variable was:

- rating
-

7.3 Train-Test Split

The dataset was divided into training and testing datasets.

The split was:

- **80% training data**
- **20% testing data**

A random state of **42** was used to ensure reproducibility.

7.4 Data Preprocessing

The numerical features were passed directly into the models.

The categorical category variable was converted into numerical representation using **One-Hot Encoding**.

A Scikit-learn pipeline was used to combine preprocessing and model training.

This approach ensures that the same preprocessing procedure is applied consistently during training and prediction.

7.5 Random Forest Regression

The first machine learning model was a Random Forest Regressor.

The model used:

300 decision trees

with a fixed random state of 42.

The Random Forest model was trained using:

- Price
- Stock count
- Description length
- Category

to predict rating.

7.6 Random Forest Results

The Random Forest model achieved:

RMSE = 1.5236

$R^2 = -0.1328$

RMSE measures the average magnitude of prediction error, with lower values representing better performance.

The negative R^2 indicates that the model does not explain the variation in ratings well using the available features.

7.7 Linear Regression

A Linear Regression model was also developed using the same features and preprocessing approach.

This provided a baseline model against which the Random Forest model could be compared.

7.8 Linear Regression Results

The Linear Regression model achieved:

RMSE = 1.5244

$R^2 = -0.1341$

The results were slightly worse than those obtained from Random Forest Regression.

7.9 Model Comparison

Model	RMSE	R^2 Score
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Random Forest Regression	1.5236	-0.1328
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Linear Regression	1.5244	-0.1341
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Random Forest achieved:

- The lower RMSE
- The higher R^2 score

Therefore, Random Forest Regression was selected as the final model.

7.10 Final Model

The final selected model is:

Random Forest Regression

Performance:

- **RMSE: 1.5236**
- **R^2 Score: -0.1328**

The trained model was saved as:

final_book_rating_regression_model.joblib
The model comparison results were saved as:
regression_model_comparison.csv

7.11 Interpretation of Model Performance

The machine learning results are an important finding of the project.

Although Random Forest performed slightly better than Linear Regression, both models produced negative R^2 values.

This indicates that:

Price, stock count, description length, and category alone are not sufficient to reliably predict book ratings in this dataset.

This result is consistent with the EDA findings, where the relationships between rating and the numerical variables were found to be very weak.

Therefore, the machine learning phase demonstrates not only model building but also the importance of evaluating whether the available data contains sufficient predictive information.

8. PHASE 5 – POWER BI DASHBOARD

8.1 Dashboard Objective

Microsoft Power BI was used to convert the cleaned dataset and analytical findings into an interactive dashboard.

The dashboard provides users with an easy way to explore:

- Book quantity
- Ratings
- Prices
- Price categories
- Book categories
- Stock levels
- Rating categories
- Detailed book information

The dashboard contains two main pages:

1. **Dashboard Overview**
2. **Book Details**

8.2 Dashboard Overview Page

The first page is named:

Dashboard Overview

The dashboard title is:

Book Market Analysis & Rating Dashboard

The dashboard contains four main KPI cards.

KPI 1 – Total Books

1,000 books

KPI 2 – Average Rating

2.92

KPI 3 – Price Range

£10.00 – £59.99

KPI 4 – Average Price

£35.07

These KPIs provide a quick summary of the catalogue.

8.3 Dashboard Visualizations

The Dashboard Overview page includes the following visualizations.

Average Price by Price Category

This visualization compares average prices across:

- Budget
- Mid-Range
- Premium

The average prices are approximately:

- Budget: **£14.80**
 - Mid-Range: **£30.06**
 - Premium: **£49.91**
-

Average Rating by Rating Category

This visualization compares ratings across:

- Low Rated
 - Average Rated
 - Highly Rated
-

Top 10 Book Categories

A Top 10 visual was created to show the most represented book categories.

The visual helps identify which categories contain the largest number of books.

Book Count by Price Range

A price-bin visualization shows how many books fall into different price ranges.

The price bins were created using a bin size of approximately £5.

Average Stock by Price Category

This visual compares average stock levels across Budget, Mid-Range, and Premium books.

Average Price by Rating

A trend-style line chart compares average price across the five rating levels.

This supports the analysis of the relationship between price and rating.

8.4 AI-Powered Visualizations

The dashboard also includes Power BI AI visuals.

Key Influencers

The Key Influencers visual analyses:

Book Rating

using explanatory variables such as:

- Price
- Stock count
- Price category
- Category

The visual displayed:

No influences found.

This result is consistent with the EDA and machine learning findings, which showed weak relationships between the available features and book rating.

Decomposition Tree

A Decomposition Tree was created with:

Price

as the analysed value.

The analysis can be broken down using:

- Category
- Rating Category

The visual is titled:

Price Breakdown by Category & Rating

This allows users to explore the composition of catalogue prices interactively.

8.5 Interactive Slicers

The dashboard includes interactive slicers for:

- Category
- Rating Category
- Numeric Rating

The rating slicer allows users to filter ratings between 1 and 5.

These slicers allow users to dynamically explore different segments of the catalogue.

8.6 Bookmarks

A bookmark named:

Reset Filters

was created.

The bookmark allows users to return the dashboard to its default filter state after applying different selections.

This improves dashboard usability.

8.7 Drill-Through

A second page named:

Book Details

was created to support drill-through analysis.

The category field was configured as the drill-through field.

Users can therefore select a category from the overview and navigate to the Book Details page to examine the corresponding books.

8.8 Book Details Page

The Book Details page contains a table displaying important book-level information, including:

- Title
- Category
- Price
- Rating
- Stock Count

The numeric fields were configured so that individual values are displayed correctly rather than being unnecessarily aggregated.

A Back button is also available to return to the main dashboard.

8.9 Synchronized Slicers

The Category and Rating Category slicers were synchronized between the dashboard pages.

This ensures that filters selected on the main dashboard can remain consistent when navigating to the Book Details page.

8.10 Row-Level Security

Row-Level Security was configured using a role named:

Premium Books

The filtering condition was based on:

price_category = Premium

When the Premium Books role was tested, the dashboard displayed:

- **Total Books: 403**
- **Average Price: £49.91**
- **Price Range: £40.11 – £59.99**

This confirms that the dashboard supports role-based filtering of the dataset.

8.11 Dashboard Outcome

The completed Power BI dashboard combines:

- KPI cards
- Category analysis
- Rating analysis
- Price analysis
- Stock analysis
- AI visuals
- Decomposition Tree
- Interactive slicers
- Bookmarks
- Drill-through
- Synchronized slicers
- Row-Level Security

This provides an interactive interface for exploring the cleaned book catalogue.

9. KEY BUSINESS AND ANALYTICAL INSIGHTS

The project produced several important insights.

9.1 Pricing

The catalogue has a broad price range from £10.00 to £59.99, with an average price of £35.07. Premium books have an average price of approximately £49.91, compared with approximately £30.06 for Mid-Range books and £14.80 for Budget books.

This clearly separates the catalogue into three meaningful price segments.

9.2 Ratings

The average rating across the dataset is 2.92.

All five rating levels are represented, with 1-star books being the largest individual rating group at 226 books.

9.3 Price Does Not Strongly Explain Rating

The correlation between price and rating is approximately 0.028.

This is extremely weak.

The average prices for different ratings are also very similar.
Therefore, a higher catalogue price does not necessarily indicate a higher rating in this dataset.

9.4 Category Differences

There are 50 categories in the catalogue.

Suspense has the highest average price at approximately £58.33.

At the same time, categories such as Erotica, Adult Fiction, and Novels achieved an average rating of 5.00 in the analysed data.

However, category-level rating comparisons should be treated carefully because category sizes are not equal.

9.5 Stock Levels

Average stock is approximately 8.59 books.

The difference in average stock between rating groups is relatively small.

The highest average stock occurs for 4-star books at approximately 8.88.

This suggests that rating does not have a meaningful relationship with stock availability in this dataset.

9.6 Machine Learning Insight

The most important machine learning finding is that the selected features have limited predictive power for book ratings.

The Random Forest model achieved an R^2 of -0.1328.

This means that the available features do not provide enough information to accurately predict rating.

This is supported by the correlation analysis and the Key Influencers visual, which also found no meaningful influences.

9.7 Dashboard Insight

The Power BI dashboard makes the analytical findings easier to explore.

Users can move from high-level KPIs to category-level analysis and then drill through to individual book details.

The interactive filtering and RLS features also demonstrate how the analysis could be presented to different users or business roles.

10. **RECOMMENDATIONS**

Based on the analysis, the following recommendations are proposed.

Recommendation 1 – Use Price Segmentation

The catalogue should continue to be analysed using Budget, Mid-Range, and Premium segments because the three categories show clear differences in average price.

This makes pricing analysis easier to communicate.

Recommendation 2 – Do Not Use Price Alone to Estimate Rating

The analysis shows an extremely weak relationship between price and rating.

Therefore, price should not be used as a standalone indicator of book quality or rating.

Recommendation 3 – Use Additional Features for Rating Prediction

The machine learning results suggest that additional predictive variables would be required to build a stronger rating prediction model.

Potential future features could include richer textual information, author-related information, publication information, review-related information, and other attributes if available.

Recommendation 4 – Monitor Category-Level Performance

Category-level price and rating analysis should be monitored because different categories show different average prices and rating patterns.

However, categories with small sample sizes should be interpreted carefully.

Recommendation 5 – Use the Power BI Dashboard for Interactive Analysis

The dashboard should be used as the main interface for exploring catalogue data because users can filter by category and rating and drill through to individual books.

Recommendation 6 – Use RLS for Role-Based Reporting

The Premium Books RLS configuration demonstrates how the dashboard can provide different users with access to specific subsets of the catalogue.

This approach could be extended to other business roles if the project were connected to a larger real-world dataset.

11. LIMITATIONS

Several limitations should be considered when interpreting the project results.

11.1 Dataset Source

The project uses the Books to Scrape website, which is a scraping practice/sandbox website. Therefore, the results should be interpreted as **catalogue analysis and a demonstration of a data science workflow**, rather than as a representation of the entire real-world book market.

11.2 No Time Variable

The dataset does not contain a date or time field.

Therefore, meaningful daily, monthly, or yearly trends cannot be analysed.

A time-series analysis was not fabricated because no appropriate time variable exists in the dataset.

11.3 Limited Predictive Features

The rating prediction model uses:

- Price
- Stock count
- Description length
- Category

These features have weak relationships with rating.

As a result, the model has limited predictive performance.

11.4 Negative R^2

Both machine learning models produced negative R^2 scores.

Random Forest achieved:

-0.1328

Linear Regression achieved:

-0.1341

This indicates that the models do not explain the variation in ratings well.

Rather than hiding this result, it is reported as an important finding of the project.

11.5 Category Sample Size

There are 50 categories, but the number of books per category varies significantly.

Some categories contain many books while others contain relatively few.

Therefore, category-level averages, especially rating averages, should be interpreted carefully.

11.6 Constant Variables

Tax was constant at zero across the dataset, while availability was standardized to a single value.

These variables therefore provide little or no useful variation for statistical analysis or machine learning.

12. CONCLUSION

This project successfully implemented an end-to-end data science workflow for book catalogue analysis.

The project began with web scraping using Python, Requests, and BeautifulSoup. A total of 1,000 book records were collected from multiple catalogue pages and detailed book pages. The resulting raw data contained 10 fields covering book information such as title, price, rating, availability, category, UPC, stock count, tax, and description.

The raw data was then cleaned and transformed using Pandas. Missing descriptions were handled, currency values were converted to numerical form, ratings were converted from words to numbers, availability was standardized, text fields were cleaned, and additional features were created.

Exploratory Data Analysis revealed that the average book price was £35.07, the average rating was 2.92, and the average stock count was 8.59. The catalogue contained 50 categories.

Premium books had substantially higher average prices than Budget and Mid-Range books. However, the relationship between price and rating was extremely weak.

For machine learning, Random Forest Regression and Linear Regression were implemented to predict book ratings. Random Forest performed slightly better, with an RMSE of 1.5236 and an R^2 of -0.1328. However, the negative R^2 demonstrates that the selected features have limited ability to predict book ratings.

The machine learning result is consistent with the EDA and Power BI Key Influencers analysis. Together, these findings demonstrate that meaningful machine learning depends not only on choosing an appropriate algorithm but also on having sufficiently informative features.

The final Power BI dashboard provides an interactive presentation of the project findings. It includes KPI cards, charts, slicers, AI visuals, a decomposition tree, bookmarks, drill-through, synchronized slicers, and Row-Level Security.

Overall, the project demonstrates the complete process of transforming raw web data into a cleaned dataset, extracting analytical insights, developing and evaluating machine learning models, and presenting the results through an interactive business intelligence dashboard.

13. PROJECT DELIVERABLES

The following project files were produced during the project.

Python/Jupyter Notebooks

1. 01_Web_Scraping.ipynb
2. 02_Data_Cleaning.ipynb
3. 03_EDA.ipynb
4. 04_Machine_Learning.ipynb

Datasets

1. raw_books_data.csv
2. cleaned_books_data.csv

Machine Learning Files

1. final_book_rating_regression_model.joblib
2. regression_model_comparison.csv

Power BI

Book_Market_Analysis_Dashboard.pbix

Final Submission

- PDF Report
- Video Demonstration
- Complete Project ZIP

14. APPENDIX A – WEB SCRAPING CODE

The following code represents the main scraping approach used in the project.

```
%pip install requests beautifulsoup4 pandas
```

```
soup = BeautifulSoup(response.text, "html.parser")  
  
print("Page title:", soup.title.text.strip())
```

```
books = soup.find_all("article", class_="product_pod")  
  
print("Books found:", len(books))
```

```
detail_response = requests.get(product_url)  
  
print("Status Code:", detail_response.status_code)
```

```
detail_soup = BeautifulSoup(detail_response.text, "html.parser")  
  
print(detail_soup.title.text.strip())
```

```
availability_detail = detail_soup.select_one("p.availability").text.strip()  
  
print("Availability:", availability_detail)
```

```
import re  
  
stock_match = re.search(r"\\(\\d+\\) available\\)", availability_detail)  
  
if stock_match:  
    stock_count = int(stock_match.group(1))  
else:  
    stock_count = 0  
  
print("Stock Count:", stock_count)
```

```
book_data = {  
    "title": title,  
    "price": price,  
    "rating": rating,  
    "availability": availability,  
    "product_url": product_url,  
    "category": category,  
    "upc": upc,  
    "stock_count": stock_count,  
    "tax": tax,  
    "description": description  
}  
  
for key, value in book_data.items():  
    print(f"{key}: {value}")
```

```
import requests  
from bs4 import BeautifulSoup  
import pandas as pd  
import time
```

```
url = "https://books.toscrape.com/"  
response = requests.get(url)  
  
print("Status Code:", response.status_code)
```

```
book = books[0]  
  
title = book.h3.a["title"]  
price = book.find("p", class_="price_color").text.strip()  
rating = book.find("p", class_="star-rating")["class"][1]  
availability = book.find("p", class_="instock").text.strip()  
  
print("Title:", title)  
print("Price:", price)  
print("Rating:", rating)  
print("Availability:", availability)
```

```
from urllib.parse import urljoin  
  
product_url = urljoin(url, book.h3.a["href"])  
  
print("Product URL:", product_url)
```

```
category = detail_soup.select("ul.breadcrumb li a")[-1].text.strip()  
  
print("Category:", category)
```

```
upc = detail_soup.find("th", string="UPC").find_next_sibling("td").text.strip()  
  
print("UPC:", upc)
```

```
tax = detail_soup.find("th", string="Tax").find_next_sibling("td").text.strip()  
  
print("Tax:", tax)
```

```
description = detail_soup.select_one("#product_description")  
  
if description:  
    description = description.find_next_sibling("p").text.strip()  
else:  
    description = ""  
  
print("Description:", description)
```

```

all_books = []

page_url = "https://books.toscrape.com/catalogue/page-1.html"

while page_url:
    print("Scraping:", page_url)

    page_response = requests.get(page_url)
    page_soup = BeautifulSoup(page_response.text, "html.parser")

    books = page_soup.find_all("article", class_="product_pod")

    for book in books:
        title = book.h3.a["title"]
        price = book.find("p", class_="price_color").text.strip()
        rating = book.find("p", class_="star-rating")["class"][1]
        availability = book.find("p", class_="instock").text.strip()
        product_url = urljoin(page_url, book.h3.a["href"])

        all_books.append({
            "title": title,
            "price": price,
            "rating": rating,
            "availability": availability,
            "product_url": product_url
        })

    next_button = page_soup.select_one("li.next a")

    if next_button:
        page_url = urljoin(page_url, next_button["href"])
    else:
        page_url = None

    time.sleep(0.5)

print("Total books collected:", len(all_books))

```

```

df = pd.DataFrame(all_books)

print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

df.head()

```

```

raw_file = "raw_books_data.csv"

df.to_csv(raw_file, index=False, encoding="utf-8-sig")

print(f"Raw dataset saved successfully as: {raw_file}")
print(f"Total records: {len(df)}")

```

```

for i, book in enumerate(all_books):
    try:
        detail_response = requests.get(book["product_url"], timeout=10)
        detail_soup = BeautifulSoup(detail_response.text, "html.parser")

        # Category
        book["category"] = detail_soup.select("ul.breadcrumb li a")[-1].text.strip()

        # UPC
        book["upc"] = detail_soup.find(
            "th", string="UPC"
        ).find_next_sibling("td").text.strip()

        # Availability and stock count
        availability_detail = detail_soup.select_one(
            "p.availability"
        ).text.strip()

        book["availability"] = availability_detail

        stock_match = re.search(
            r"((\d+) available)", availability_detail
        )

        book["stock_count"] = int(stock_match.group(1)) if stock_match else 0

        # Tax
        book["tax"] = detail_soup.find(
            "th", string="Tax"
        ).find_next_sibling("td").text.strip()

        # Description
        description_tag = detail_soup.select_one("#product_description")

        if description_tag:
            book["description"] = description_tag.find_next_sibling("p").text.strip()
        else:
            book["description"] = ""

    except Exception as e:
        print(f"Error at book {i + 1}: {e}")

        if (i + 1) % 100 == 0:
            print(f"{i + 1} books processed")

        time.sleep(0.2)

print("Detailed scraping completed!")

```

```

print("Column names:")
print(df.columns.tolist())

print("\nMissing values:")
print(df.isnull().sum())

```


15. APPENDIX B – DASHBOARD SCREENSHOTS

The following screenshots should be inserted into this section of the final Word report.

Figure 1 – Power BI Dashboard Overview



Figure 1: Book Market Analysis & Rating Dashboard – Overview page

The overview page presents the main KPIs, price analysis, rating analysis, category analysis, stock analysis, AI visuals, and interactive filters.

Figure 2 – Power BI Book Details Page

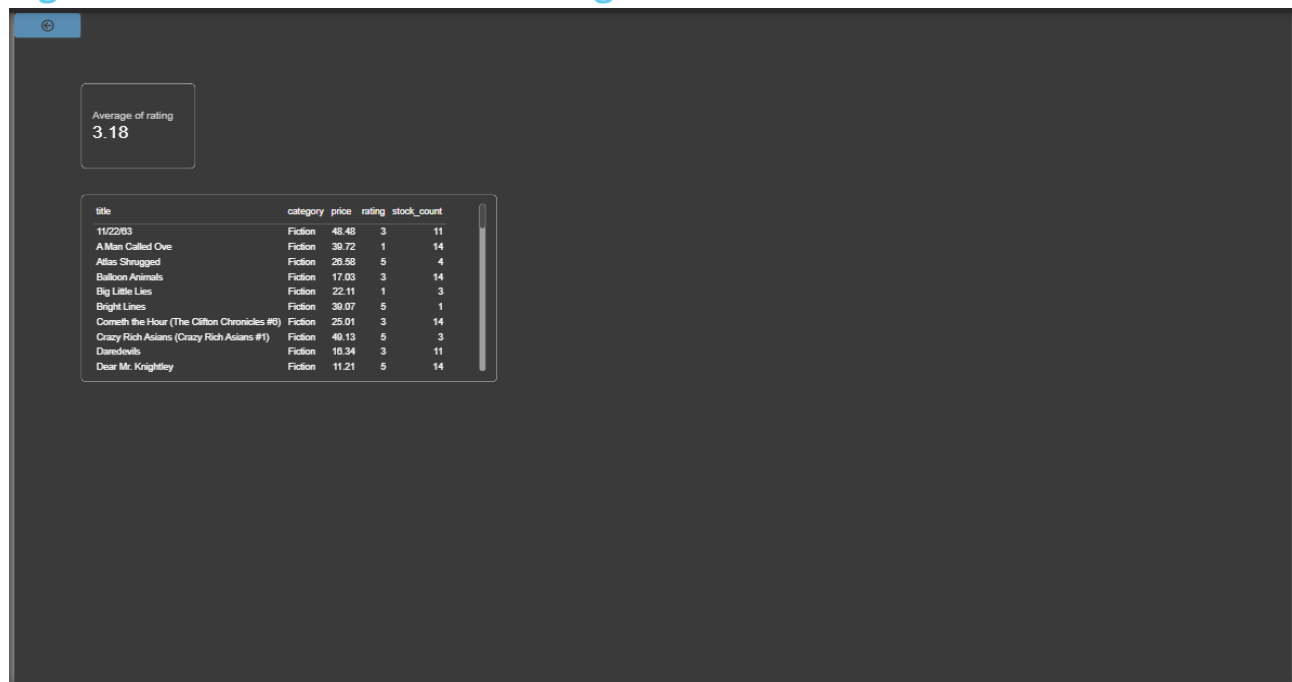


Figure 2: Book Details page showing drill-through book information

The Book Details page allows users to examine individual book records after applying category-based drill-through filtering.

Figure 3 – Power BI RLS Test



Figure 3: Row-Level Security test for Premium Books

The Premium Books role filters the dataset to books belonging to the Premium price category. The test produced:

- Total Books: 403
- Average Price: £49.91
- Price Range: £40.11–£59.99

Figure 4 – Machine Learning Model Comparison

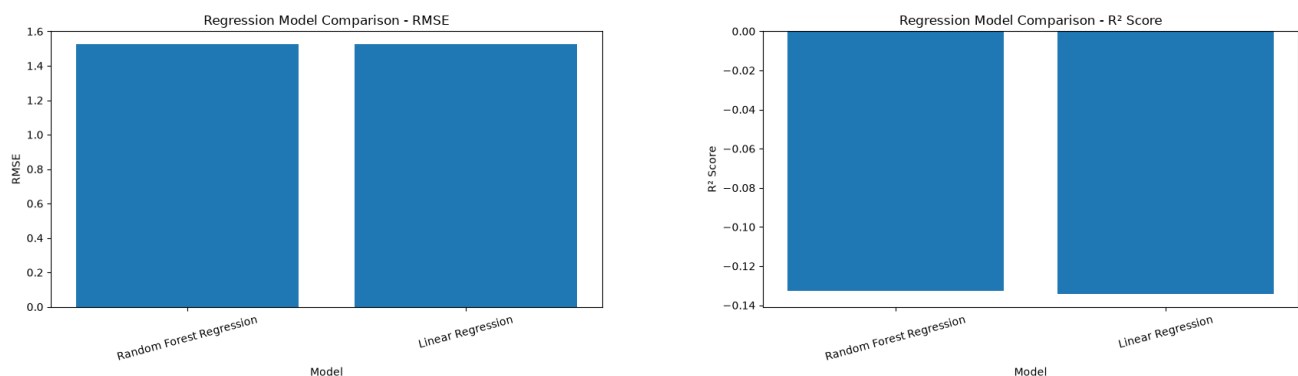


Figure 4: Comparison of Random Forest Regression and Linear Regression

The comparison demonstrates that Random Forest achieved slightly better performance than Linear Regression.

Figure 5 – EDA Visualizations

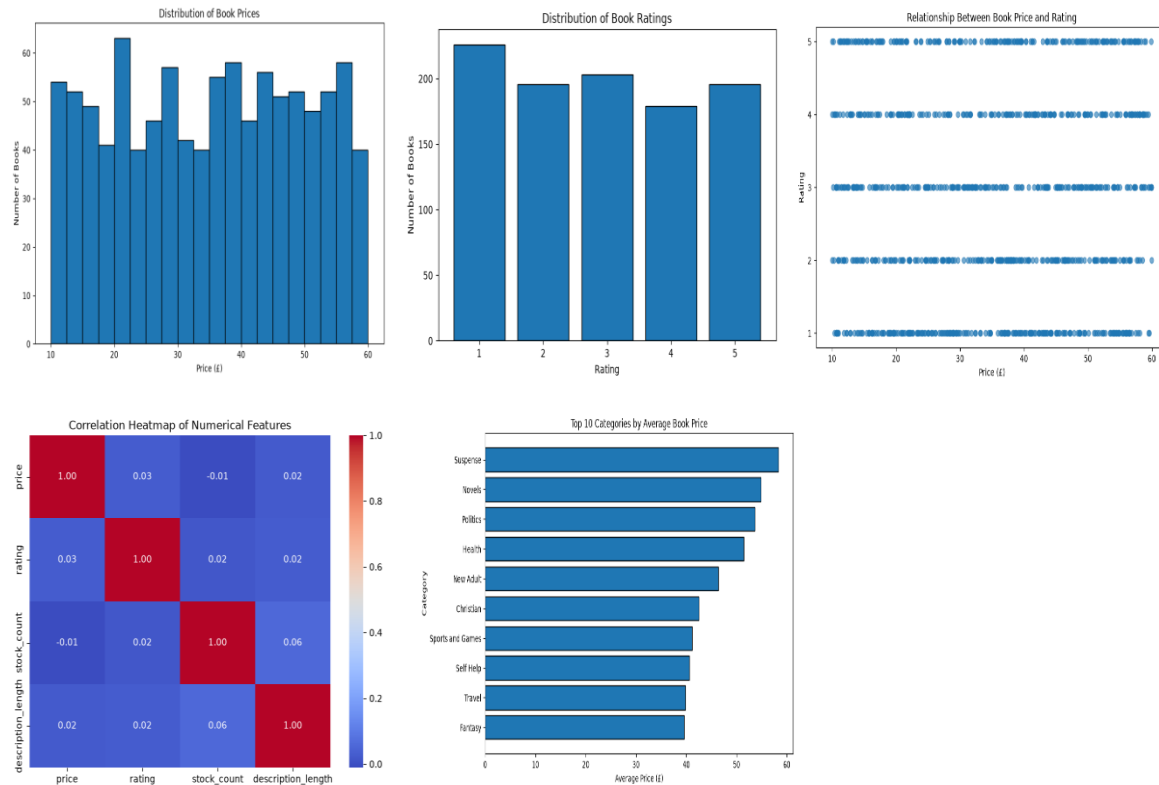


Figure 5: Selected Exploratory Data Analysis visualizations

16. APPENDIX C – VIDEO DEMONSTRATION

Video Demonstration

The project video demonstrates:

1. Project objective
2. Web scraping process
3. Raw dataset creation
4. Data cleaning
5. Exploratory data analysis
6. Machine learning model development
7. Model evaluation
8. Power BI dashboard
9. Interactive filtering
10. Drill-through
11. Row-Level Security
12. Final project findings

Video Link

<https://drive.google.com/file/d/1yyEpXRxL36BjDZrwCNG0cXCo3pBJV27j/view?usp=sharing>

17. FINAL PROJECT SUMMARY

Phase	Completed Work	Main Output
Phase 1	Web scraping and data collection	raw_books_data.csv
Phase 2	Data cleaning and feature engineering	cleaned_books_data.csv
Phase 3	EDA and visualization	EDA findings and charts
Phase 4	Regression modelling	.joblib model + comparison CSV
Phase 5	Interactive Power BI dashboard	.pbix dashboard
Phase 6	Final documentation and presentation	PDF + Video + ZIP

FINAL STATEMENT

This project demonstrates an end-to-end application of data science techniques, beginning with web data acquisition and continuing through data preparation, exploratory analysis, machine learning, and interactive dashboard development.

The project successfully collected and processed 1,000 book records, identified important catalogue patterns, evaluated two regression models, selected Random Forest as the better-performing model, and developed an interactive Power BI dashboard.

Although the rating prediction model had limited predictive performance, this result itself provided an important analytical conclusion: the available catalogue features do not contain enough information to reliably predict book ratings. The combination of EDA, machine learning evaluation, and Power BI analysis provides a consistent view of this limitation.

The completed project therefore demonstrates not only technical implementation but also the ability to interpret results, identify limitations, and communicate findings through a professional analytical report and interactive dashboard.